AI will accelerate biotech development as regulation lags

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AI is rapidly lowering technical barriers in biotech for both rogue actors and industry players, but the most dramatic risks and potential benefits of AI-enabled biotech, including AI-designed pathogens with pandemic potential and a sharp reduction in drug development timelines, remain unlikely within the next year.

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Countries hosting frontier AI biological model development will prioritize competitiveness over slower model development or guardrails designed to reduce biotech-related risks, while governments will implement regulations targeting feasible non-AI chokepoints unevenly, if at all.

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As the likelihood of AI-enabled biotech breakthroughs rises for legitimate and malicious purposes alike, AI companies will become more directly involved in biotech, health data will become more commodified and restricted, and defense strategies will shift from prevention toward detection and response preparedness.

(This report is the first in a series on the intersection of AI and biotech.)

The risks and opportunities involving AI and biotech are widely misunderstood. AI-designed novel bioweapons that threaten a new pandemic are unlikely to pose a threat within the next year, but AI is rapidly lowering barriers to would-be bioterrorists bent on causing disruption. Similarly, AI will not instantly solve the notoriously long and difficult path to developing lifesaving medications but will enable increasingly frequent breakthroughs in the coming months by nimble, flexible companies.

Biological AI models that can design a possible new human virus will likely emerge within the next year. Foundational models trained on genetic data and the resulting protein shapes and functions already exist. Scientists have used one of these models to design and create viable bacteriophages—viruses that infect bacteria—that could reproduce and kill their host, illustrating proof-of-concept that the most advanced AI tools can design whole new viral genomes. No public example has yet shown scientists training such models on human virus genomes, but researchers will almost certainly keep testing the boundaries of what is possible and may not disclose all such work.

However, a newly designed virus would not immediately become a deadly biothreat. In the proof-of-concept work with bacteriophages, only a small percentage of predicted viral genomes were viable, and validation of these sequences required substantial lab work to create and test each one.

The full potential of frontier AI models for drug design may take years to become clear. Rapid progress in AI biological models has produced promising breakthroughs. Protein structure modeling can allow the design of small molecules that fit perfectly into molecular targets, while scientists can now design entirely new proteins for specific purposes. However, many of these candidates may not perform as well in complex biological systems; bringing a promising molecule through lab-based and human testing can take years.

Implementing AI-designed biological products will be easier for legitimate users and state-backed groups than for smaller rogue groups. AI-enhanced biotech is inherently dual-use: The same inputs and infrastructure can support either a biological therapy or a dangerous pathogen. Access to the physical infrastructure needed to create an AI-designed product, and the expertise required to operate it, remains a much greater barrier for lone actors than for companies and state groups (see figure).

Specialized inputs will be a key chokepoint; access biological laboratory facilities and expertise will remain the highest barrier for lone wolf actors or small terrorist groups. Synthetic DNA needed to alter an existing genome or build a new one is the most specialized input.As a result, regulators have focused most heavily on it, but those efforts have been limited, and current frameworks are voluntary.

Both pharma companies and state bioweapons or defense programs will have access to the physical infrastructure and expertise needed to implement genetic engineering techniques and validate potential products. This step is the highest barrier for rogue actors but is falling quickly. LLM models can rapidly integrate publicly available text-based information to help scientists troubleshoot laboratory procedures. Agentic biological systems further reduce the necessary expertise by suggesting specific experiments or procedures to achieve a stated goal.

Automated or cloud labs will introduce additional risks and opportunities.The rise of cloud labs—largely automated labs that can perform wet lab work from experimental protocols that scientists submit remotely—has opened a new frontier for scientific discovery. Promising ideas can be explored without investment in physical infrastructure, and the automation of repetitive tasks through a robotic interface can dramatically cut costs and time. Most cloud labs still rely on human oversight, but greater automation may increase the risk that users could submit and execute dangerous work.

Minimal and uneven regulation of potentially dangerous lab work increases the risk of misuse and accidental release of pathogens. Biological lab facilities designated as biosafety level 3 or 4 (BSL-3 or BSL-4), indicating that they work with dangerous pathogens requiring additional safety precautions, have proliferated worldwide in recent years, partly in response to the Covid pandemic. Over 90% of countries with at least one BSL-3 lab lack mandated training or regulations for the oversight of potential dual-use research. Large cloud labs operate in China, the US, and Europe, and face limited regulation, if any. A pneumonia death in late September, reportedly linked to plague, of a lab worker in Russia is only the most recent reminder of the potentially disastrous consequences of a lab leak. A pathogen doesn’t have to be intentionally weaponized to cause substantial harm (see figure).

The biggest AI-enabled biothreats within the next year will come from available models and known pathogens. Only the most sophisticated state-supported labs and other large research groups will be capable of leveraging frontier biological models for a bioweapons program within the next 12-18 months. Such large groups will be far less likely to intentionally release a pathogen. On the other hand, smaller rogue states or lone-wolf actors will be able to use available LLM models to plan attacks using

known pathogens. Multiple red-team exercises have shown that users can persuade LLM models to provide information to plan a terrorist attack. In September, Anthropic announced that several users had already tried to query its latest model on potentially dual-use pathogen research.

Most large multinational pharma companies are embracing AI but will struggle to incorporate it into legacy R&D infrastructure, while AI companies will become more direct participants in biotech. Implementation has been focused on machine learning algorithms to help narrow drug targets, design molecules, or optimize clinical trial design. In addition to the challenges incorporating innovative AI approaches into existing R&D pipelines, pharma companies will struggle to compete with tech companies for top AI talent.

Anthropic's recently announced biology wet lab is the most well-developed example of in-house capacity building. The company has pledged not to compete with established pharma companies, but rather use its lab as a discovery engine, pursuing proof-of-concept projects like developing therapies for neglected diseases. Other large AI labs have spin-off or sister biotech companies that are more flexible than traditional pharma companies in leveraging the technology.

Rapid advances in AI and biotech also introduce a new dynamic in pharma industrial espionage. One company's breakthrough may become easier for competitors to reverse engineer. Companies have already grown more guarded about development plans and early-stage research, a trend that will complicate venture capital efforts.

Without a major shift in trajectory, companies and research groups will develop AI models capable of designing genomes for new viruses and pathogens. Given the rapid dissemination of even restricted models, these capabilities will eventually reach the hands of bad actors. No government has both the willingness and ability to enforce guardrails on AI development or deployment. In the US and China, the focus on AI and biotech competition will disincentivize burdensome regulations while concern about other countries' bioweapons capabilities will make biological model leadership a defense priority.

In both the US and China, any slowdown in AI development is highly unlikely for the foreseeable future. Collective efforts by top labs to pace model development risk being investigated as anticompetitive behavior, as there is no support for an antitrust waiver to explicitly allow such coordination (see: Mutual AI restraint requires antitrust waiver no one supports, 22 September 2026). In a recent executive order banning federal funding for laboratory work that could lead to more dangerous pathogens, AI-based in silico work designing such pathogens was explicitly permitted (see: Healthcare Lifeline: China will blame scientists for trial deaths rather than reform system, 06 August 2026).

China's leadership, similarly, sees AI competitiveness—especially in biotech—as a key priority. It views calls for pacing frontier model development as an effort to curb its AI ambitions. The recent summit between presidents Donald Trump and Xi Jinping produced only limited agreement to communicate about AI risks and no commitment to alter development trajectories. China's recently released five-year plan for pharma calls for greater utilization of AI to drive drug development and medtech growth, while the government has resisted calls for deeper collaboration on biothreats because it remains confident in its own defense capabilities.

For both the US and China, the dual-use nature of biotech work and the interdependencies between the two countries' industries make it highly difficult to design any import, export, or investment restrictions that address biotech security risks without harming economic interests.

The EU, for its part, has already passed and begun implementing its AI Act, the most comprehensive AI legislation of any major jurisdiction, but has delayed implementation of high-risk applications, including many healthcare uses, because of competitiveness concerns. Both the US and China, where most frontier work is concentrated, would also resist EU efforts to impose meaningful constraints on development.

AI companies' attempts to prevent misuse and abuse of their models will be insufficient to prevent bad actors from leveraging the technology. Anthropic's recent disclosure that several users had tried to access potentially dual-use biological design information underscores both the risk and companies' awareness of the issue. Companies are working to prevent models from providing harmful information, but red team efforts have consistently shown that a motivated user can find workarounds to such limits. Further, open-source models are rapidly catching up to frontier models, and once released, their use will be impossible to monitor. Open biological models may lag the frontier by 6-12 months, but LLM models that can facilitate lab work are already widely available.

The most feasible regulatory target will be the steps after AI-enabled design work, but comprehensive global regulations are highly unlikely. In the US, the administration scrapped synthetic DNA regulations imposed under Joe Biden along with the rest of the previous president's executive order on AI when Trump began his second term. New guidance on synthetic DNA regulation called for in Trump's 25 May 2025 Executive Order, "Improving the Safety and Security of Biological Research," is now nearly a year overdue. In this regulatory void, many of the largest synthetic DNA companies have agreed to a voluntary framework intended to prevent the production of potentially harmful sequences.

Synthetic DNA screening is difficult because users can slightly alter dangerous sequences or order them in pieces from different providers, and AI programs can help them do so. Advanced algorithms exist that can flag potentially harmful synthetic DNA sequences even if they have been adjusted to avoid detection, but few, if any, providers employ such methods. Know your customer requirements are more widely employed by many large synthetic nucleic acid providers, but adoption of voluntary biosafety frameworks is not universal. The probability of globally mandated screening remains very low.

For healthcare companies generally, regulatory risk from AI-enabled biological design is unlikely to become material in the near term. Short of a major misuse or accidental release, leading governments are likely to prioritize the economic and scientific advantages of AI-enabled biotech over international rules that meaningfully slow development.

The exception is firms operating at identifiable biological chokepoints. DNA/RNA synthesis providers, lab equipment manufacturers, and biotech platform providers are likely to face the earliest targeted requirements. Those could include sequence screening, biosecurity modernization requirements, and restrictions on cross-border access to sensitive data. Governments could adopt such measures as part of efforts to secure domestic biotech capacity and limit adversaries' access, without slowing AI development overall.

The increasing commodification of health data will raise compliance and cybersecurity risks for companies. The US, the EU, and China have all implemented data protection regulations of varying scope, and companies must carefully assess how operations across borders will comply with these different regulations. Demand for diverse health data will also raise the risks that healthcare companies and providers will be targeted by cyberattackers seeking to access sensitive data.

Defense strategies will have to reckon with the impossibility of preventing or deterring bioattacks. No regulatory approach can completely prevent determined actors from leveraging AI to design and create

a dangerous bioweapon. The range of actors now with access to these capabilities also reduces the effectiveness of traditional deterrence: When non-state actors are increasingly likely to launch hard-to-trace attacks, the threat of a military response carries less weight.

The specter of a weaponized pathogen may force governments to make investments in biosurveillance that failed to materialize after the Covid pandemic.In the first few years after Covid emerged, there was substantial discussion of the need for air, wastewater, and early clinical monitoring systems to detect emerging pathogens. Some progress followed, but most initiatives lost momentum. Interest has now revived in the form of advanced AI-powered detection devices and systems, some of which are already working with national security and defense partners.

In the US, capacity building will be an uphill battle following recent turnover at health agencies. Ongoing measles outbreaks and a prolonged cyclospora outbreak over the summer demonstrated the limited capacity of the current administration's health agencies, while pandemic preparedness efforts have been largely abandoned.

The potential need for rapid biothreat response is the best rationale for biotech competitiveness as a defense priority. In recent years, policymakers have identified biotech as a matter of national security but have at times struggled to make the case for prioritizing economic competitiveness or manufacturing capabilities. Planning for a possible biothreat that would require rapid mobilization of domestic expertise and infrastructure offers perhaps the most convincing rationale for incentivizing domestic R&D and manufacturing capabilities.

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